

SAGAR-SAKSHI · EVIDENCE CARD

Smart India Hackathon 2026 | Team Tantragyan (SIH26143) | Run ID: run_kerala_20260924_074608

EVIDENCE GRADE	GRADE None
HEADLINE ASSESSMENT	Insufficient evidence
AIS DATA STATUS	SYNTHETIC (Validated replay log)
MANIFEST HASH (SHA-256)	e12aecc8246e7b8ae24c5ef58bc9393a55e415054bed857b29eeb471fcec1728

Candidate Hypotheses & Posterior Probabilities P(H|E)

Hypothesis / Source	Posterior P(H E)	Drift Fit	Head Prox.	Behavior
Null / Unknown Source (U)	0.9941 (99.4%)	N/A	N/A	N/A
DARK_sar_ship_001	0.0007 (0.1%)	0.05	0.05	0.30
DARK_sar_ship_002	0.0007 (0.1%)	0.05	0.05	0.30
DARK_sar_ship_003	0.0007 (0.1%)	0.05	0.05	0.30
DARK_sar_ship_004	0.0007 (0.1%)	0.05	0.05	0.30
DARK_sar_ship_005	0.0007 (0.1%)	0.05	0.05	0.30
DARK_sar_ship_006	0.0007 (0.1%)	0.05	0.05	0.30
DARK_sar_ship_007	0.0007 (0.1%)	0.05	0.05	0.30
DARK_sar_ship_008	0.0007 (0.1%)	0.05	0.05	0.30
MSC ELSA 3	0.0002 (0.0%)	0.02	0.02	0.30
OCEAN PRIDE	0.0002 (0.0%)	0.02	0.02	0.30

Exculpatory Evidence: Excluded Vessels

Vessel ID	Exclusion Reason
KERALA STAR	Track never entered backward reachability envelope (closest approach: 15.0 nm)
MALABAR QUEEN	Track never entered backward reachability envelope (closest approach: 6.9 nm)

Operational Caveats & Legal Disclaimers

- Non-Accusatory Standard:** SAGAR-SAKSHI produces investigative leads based on multi-sensor evidence consistency. It does not establish legal liability or criminal culpability.
- Detectability Mask:** SAR oil detection depends on Bragg scattering in 3-10 m/s wind regimes.
- Radar Limits:** Sentinel-1 C-band SAR resolution limits reliable vessel detection to craft >= 20m.